

Subject Code	AF5323
Subject Title	Fixed Income Securities
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: Investments (AF5344)
Objectives	This course is concerned with fixed income securities and interest rate risk management. It will introduce tools used to explore the theoretical and empirical aspects of fixed income securities and their derivatives. It contributes to the achievement of the MoF programme outcomes by enabling students to understand and explain real life issues related to fixed income securities (Outcome 2), and apply relevant concepts and tools to solve problems on fixed income investment (Outcome 2).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand and explain the issues in pricing, hedging, and arbitrage in the fixed income securities markets. - Evaluate various types of fixed income products and analyze their potential risk and return. - Apply theories and concepts learned and appreciate fixed income investment decisions. - Understand and explain the recent developments and issues of the fixed income markets.
Subject Synopsis/ Indicative Syllabus	The Basic Products Bond price arithmetic. Treasury bills, notes, bonds and strips. Organization of government bond markets. Spot rates, par rates and forward rates. Constructing zero curves. Risk Management Measures of price sensitivity. Simple hedging strategies using fixed income derivatives. Eurodollar futures. Bond futures. Interest rate swaps. Pricing Interest Rate Claims Theories of the term structure. Arbitrage free pricing. Corporate Securities and Credit Risk Corporate bonds and credit risk. Credit derivatives. Mortgages and Their Derivatives Mortgages and mortgage backed securities. Prepayment risk. Bonds with Embedded Options Basic pricing principles. Static spread and option-adjusted spread. Negative convexity. Effective duration and convexity. FinTech Applications of blockchain, artificial intelligence/machine learning, big data analytics or cloud computing in bond markets.

Teaching/Learning Methodology	Lectures and seminars will be conducted on the topics of the syllabus. Lecture time will be used flexibly for discussing key concepts and their applications with students and carrying out other learning activities with them. Such activities include group discussions and student presentations of their work (to develop students' critical thinking, analytical skills, teamwork, and communication skills). To maximize benefits, students are encouraged to share their views and experiences actively with their lecturer and classmates.																																																														
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr><tr><td>1. Mid-term Test</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Individual Essay</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Group Presentation</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>4. Participation</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>5. Final Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Mid-term Test – It is a closed-book test to cover the intended subject learning outcomes. Individual Essay – Students have to do a case study or write on a topic about fixed income securities or markets so as to test their abilities to apply concepts taught. Group Presentation – Students are required to work on a group basis and present an analysis of a current issue about fixed income securities or markets. They have to demonstrate their understanding of concepts taught and their abilities to explain the recent development of the markets. Participation – Students have to actively discuss questions presented to them in classes to show their understanding of concepts taught and their abilities to apply relevant tools to analyze fixed income securities products. Final Exam – It is a 3-hour closed-book exam with compulsory problem-solving type and essay type questions to test students' understanding of and abilities to apply all concepts taught. Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers. To reflect the significant technology content in this subject, 10% of the overall weighting of this subject is based on individual assessment concerning technology-related knowledge.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d			1. Mid-term Test	10%	✓	✓	✓				2. Individual Essay	15%	✓	✓	✓	✓			3. Group Presentation	15%	✓	✓	✓	✓			4. Participation	10%	✓	✓	✓	✓			5. Final Examination	50%	✓	✓	✓	✓			Total	100 %						
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Student Study Effort Expected	Class contact:	
	▪ Lectures / Seminars	39 Hrs.
	Other student study effort:	
	▪ Preparing for classes and reviewing course materials.	39 Hrs.
	▪ Writing individual essay	10 Hrs.
	▪ Preparing for group presentation	10 Hrs.
	▪ Preparing for mid-term test and final exam	20 Hrs.
	Total student study effort	118 Hrs.
Reading List and References	<u>Textbook</u> Fabozzi, F., <i>Bond Markets, Analysis, and Strategies</i>, 8th edition, Pearson, 2013. <u>References</u> Supplementary readings from academic/professional journals and websites.	